Democratizing LoRa Weak Signal Demodulation

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Abstract

This paper democratizes LoRa weak signal demodulation from both the methodological and experimental perspectives. Methodologically, by drawing inspiration from LoRa's CSS modulation principle, this paper proposes DeLoRa, the first lightweight neural-enhanced method for LoRa weak signal demodulation which can be executed on LoRa nodes possessing far more limited hardware resources than LoRa gateways. Experimentally, this paper proposes MCU-SDR, the first hardware testbed that matches the stringent resource constraints of LoRa nodes, enabling LoRa researchers to carry out in-situ evaluation of demodulation methods for LoRa nodes. Our extensive evaluation in indoor and outdoor environments shows that DeLoRa operates effectively (i.e., ensures <10% symbol error rate) at SNR level as low as -18.8 dB, achieving a **3.3 dB SNR gain over SOTA demodulation methods that can run on LoRa nodes.**

1 INTRODUCTION

Motivation. LoRa, a low-power long-range networking technology, stands as a cornerstone of global IoT deployments [1]. **Downlink** communication from LoRa gateway **to LoRa node** is indispensable for basic functions [2], such as device activation, adaptive data rate, and firmware updating. However, LoRa nodes usually encounter the critical challenge of demodulating weak signals (SNR <-15 dB). This stems from the fact that radio regulations impose limits on the transmission power of LoRa gateways [3], and the need for LoRa nodes to operate in complex environments with severe signal blockage and interference (e.g., underground mines [4], wall-surrounded residential buildings [5], industrial facilities abundant with co-frequency interference [6]).

Existing LoRa weak signal demodulation methods [7–15], while effective for LoRa gateways, are inapplicable to resource-constrained LoRa nodes. Specifically, the hardware resources available on a LoRa node for demodulation primarily come from its LoRa transceiver chip with only MOPS-level computing power and KB-level memory, but those required by [7–15] are at least TOPS- and MB-level.

Even though one manages to develop lightweight methods for weak signal demodulation on LoRa nodes, the lack of a suitable testbed still hinders the in-situ evaluation. Both commercial LoRa nodes and existing Software-Defined Radios (SDRs), in particular, are unsuitable to serve as testbeds for LoRa node demodulation. On the one hand, commercial LoRa nodes rely on proprietary LoRa chips [16] that use closed-source I/Q signal acquisition and do not support self-defined algorithm deployment. On the other hand, existing SDRs [17–19] are unable to emulate the in-situ computing behaviors of LoRa nodes, as they either require an external PC or on-board FPGA as the computing hardware, which are architecturally far more advantageous than that of LoRa nodes in memory bandwidth and instruction parallelism.

Objective. Driven by the aforementioned motivation, this paper aims to *democratize LoRa weak signal demodulation* from both the methodological and experimental perspectives. That is, this paper aims to (i) develop a lightweight and effective method for LoRa weak signal demodulation which can be executed on ubiquitous LoRa nodes possessing much more limited hardware resources than LoRa gateways, and (ii) aims to build a low-cost hardware-matched testbed that enables LoRa researchers to conveniently record baseband I/Q signals and carry out in-situ evaluation of downlink demodulation methods. To realize such objective, this paper builds a *LoRa node demodulation testbed*, called MCU-SDR, as well as a *lightweight LoRa demodulation method*, namely DeLoRa, whose main design is CSSNet, a chirp spread spectrum (CSS)-informed neural demodulator. In what follows, we present the key designs related to MCU-SDR and CSSNet.

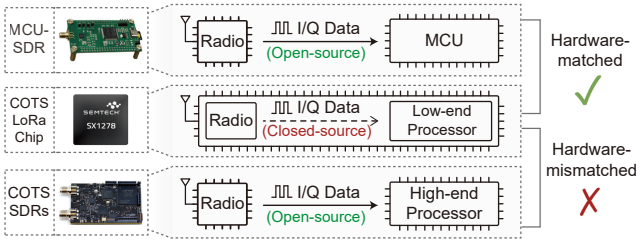


Figure 1: MCU-SDR, the proposed LoRa node demodulation testbed, has comparable computing resources to commercial LoRa chips and supports I/Q acquisition.

LoRa Node Demodulation Testbed: MCU-SDR. MCU-SDR comprises two core components: an MCU and a radio chip, as shown in Fig. 1. The radio chip transfers I/Q signals to the MCU, which runs the demodulation software to be evaluated. Although such design differs from that of commercial LoRa nodes whose radio chips handle demodulation, MCU-SDR has comparable computing resources to LoRa chips and meanwhile supports flexible deployment of demodulation algorithms.

Radio Chip: SX1255. MCU-SDR adopts SX1255 [20] as its radio chip, which architecturally inherits the same RF front-end design as most COTS LoRa chips [16]. Importantly, SX1255 transfers baseband I/Q signals via the Inter-IC Sound (I2S) protocol that is natively supported by most MCUs. In fact, most existing radio chips either operate in frequency bands incompatible with LoRa [21], or rely on high-speed protocols (e.g., LVDS, USB 2.0) for I/Q signal transfer [22], whose data rates (>400 Mbps) far exceed the GPIO capabilities of MCUs (<100 Mbps). It is worth noting that building MCU-SDR requires reverse-engineering and re-implementing SX1255 at the register level, as no open-source driver or reference code is available for its digital baseband interface. We developed the firmware from scratch to configure all of its 19 registers with each containing 8 bits, enabling flexible control over a series of parameters (e.g., I/Q sampling rate, I/Q transfer mode, decimation factor).

CPU-less I/Q Streaming. Due to the lack of open-source implementations for I/Q streaming on MCUs, we developed a highly optimized MCU configuration for this task. The MCU is configured

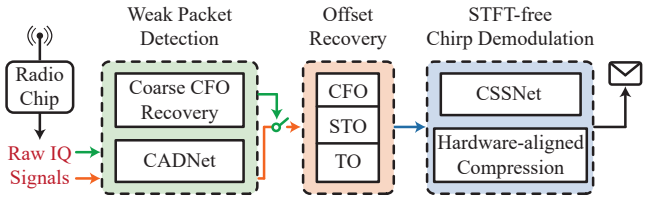


Figure 2: Architecture of the proposed DeLoRa.

to operate in direct memory access (DMA) mode, where the DMA module leverages a dedicated bus to directly receive I/Q signals from SX1255 via I2S, liberating the CPU of MCU from managing low-level I/Q streaming and enabling it to focus entirely on executing demodulation algorithms. Additionally, we implement a double-buffer mechanism: as soon as the DMA fills up one buffer, it sends an interrupt to the CPU, and then the CPU processes the I/Q signal data in that buffer, while the DMA transfers the incoming I/Q stream into another buffer. Compared to the single-buffer design, this mechanism prevents newly arriving I/Q stream data from overwriting buffered data during CPU processing.

CSS-informed Neural Demodulator: CSSNet. Recent works have successfully leveraged the feature extraction power of neural networks to enhance LoRa weak signal demodulation [10–12]. Instead of simply treating it as a generic time-series task, we draw inspiration from LoRa's CSS modulation principle and propose CSSNet as DeLoRa's neural demodulator, enabling lightweight extraction of discriminative chirp features from noisy LoRa I/Q signals.

Lightweight-oriented Neural Architecture. Given that typical MCUs operate under strict memory constraints (<1 MB) and support limited operators [23, 24], CSSNet was designed inherently to prioritize this lightweight requirement. Specifically, CSSNet adopts a lightweight fully convolutional neural architecture to extract temporal features directly from I/Q signals, whose model size is only 4.6 MB, suitable for MCU deployment after compression. In contrast, SOTA neural LoRa demodulators universally employ an encoder-decoder architecture based on CNN+LSTM [10], CNN+GRU [11], or transformer models [12], that occupy tens of MB and require complex operators, rendering them unsuitable for MCUs.

CSS-informed Convolution. The lightweightness of CSSNet, though enabling MCU deployment, limits its representation capability [25]. We thus make LoRa-specific designs on the atomic convolution operation to ensure its performance for demodulating LoRa weak signals. Specifically, LoRa's CSS modulation principle defines that the chirp of each distinct symbol exhibits a unique linear frequency variation pattern over time. Inspired by this fact, we propose CSS-informed convolution with a multi-scale receptive field spanning 1/12 to 1/128 of the chirp length across layers, enabling filters to robustly capture each chirp's linear frequency-variation patterns at various granularities. Moreover, CSS specifies that LoRa symbols are modulated by cyclically shifting the initial frequency of a base upchirp, and thus the boundaries of each chirp are crucial for demodulation. We accordingly adopt a convolution extension strategy that intentionally shifts the convolutional starting position before the first sample and the ending position after the last sample, enhancing the perception of frequency changes at chirp boundaries.

Fast Weak Preamble Detection. Before applying CSSNet, a LoRa node has to detect LoRa preambles. Clearly, different from LoRa demodulation that maps I/Q signal to one of multiple symbol classes, preamble detection is in fact a binary classification task. Furthermore, LoRa preamble follows a fixed symbol pattern. Such two facts make LoRa preamble detection simpler than demodulation. Leveraging this, this paper proposes a minimalist preamble detection model which contains only two CSS-informed convolutional layers and processes directly on half of the in-phase signal without FFT computation, achieving low-latency LoRa weak preamble detection.

Contributions. We present the key contributions of this paper in terms of methodology, testbed, and performance.

- *Methodology.* This paper proposes DeLoRa, the first lightweight demodulation method for resource-constrained LoRa nodes operating at extremely low SNR. It features a CSS-informed neural demodulator and detector that extract discriminative chirp features from noisy I/Q signals.
- *Testbed.* This paper introduces MCU-SDR, the first testbed that enables in-situ evaluation of LoRa node demodulation methods. It features custom register-level firmware and an optimized MCU configuration for CPU-less I/Q streaming.
- *Performance.* This paper conducts comprehensive experiments in both indoor and outdoor environments. Experiment results show that DeLoRa operates effectively at SNR levels as low as -18.8 dB, achieving a 3.3 dB SNR gain over SOTA demodulation methods that can run on LoRa nodes.

2 DeLoRa OVERVIEW

Fig. 2 shows the architecture of proposed DeLoRa. In the following, we give an overview of DeLoRa's three components.

Weak Preamble Detection. Prior to demodulation, the weak preamble detection controls I/Q signals streaming to the following modules. Due to the carrier frequency offset (CFO) problem between devices, even identical chirps exhibit phase discrepancies. We employ coarse CFO recovery to preliminarily enhance the signal, followed by the proposed CADNet to achieve robust LoRa preamble detection with ultra-low latency even under extremely low SNR.

Offset Recovery. After capturing raw LoRa I/Q signals, the offset recovery module performs signal synchronization. Specifically, we employ an efficient phase search method to identify and eliminate CFO. Leveraging higher sampling rates and a robust demodulator introduced below, we eliminate sampling time offset (STO) and timing offset (TO).

STFT-free Chirp Demodulation. After the lightweight-oriented design, CSSNet undergoes hardware-aligned compression, reaching a compact size and dense computing graph that are suitable for LoRa nodes. Synchronized LoRa I/Q signals are then demodulated end-to-end by CSSNet, avoiding computationally intensive Short-Time Fourier Transform (STFT) operations.

To align with technical logic and significance, we introduce DeLoRa in the following order: the *CSS-informed neural demodulator* (§ 3), the *weak preamble detection* technique (§ 4), and the *efficient offset recovery* technique (§ 5).

3 CSS-INFORMED NEURAL LoRa DEMODULATOR

The intuition behind the CSS-informed neural demodulator, is to draw inspiration from the CSS modulation principle for lightweight yet reliable extraction of discriminative chirp features from noisy

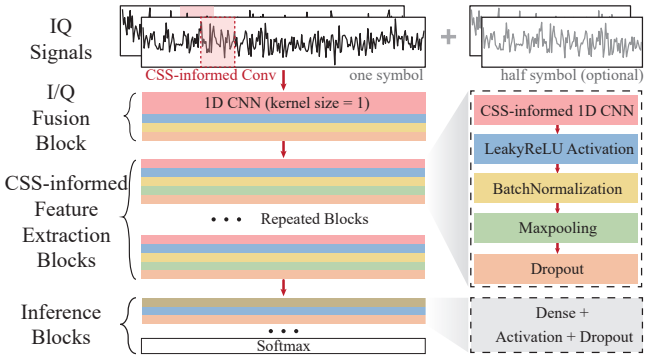


Figure 3: Model structure of the proposed CSSNet.

LoRa I/Q signals. Specifically, such technique consists of *lightweight-oriented processing* (§ 3.1), *CSS-informed model design* (§ 3.2), and *hardware-aligned model compression* (§ 3.3) described as follows.

3.1 Lightweight-oriented Processing

Fully-conv Architecture. Given that MCUs typically possess limited memory space (<1 MB), embedded neural networks should be lightweight enough for successful deployment [23]. We emphasize such lightweight requirement and carefully design CSSNet with a suitable architecture while maintaining competitive accuracy. Specifically, CSSNet features an MCU-compatible fully convolutional architecture designed for the lightweight extraction of temporal features from LoRa I/Q signals. In contrast, leading neural LoRa demodulators rely on either CNN+LSTM/GRU [10, 11] or transformer [12] models as the encoder-decoder architecture for processing LoRa signals. These models employ deep hierarchies of layers and parameter-rich mechanisms like multi-head attention, occupying tens of MB in size, thereby being difficult to be deployed on LoRa nodes.

STFT-free Processing. Existing neural LoRa demodulators rely on STFT operation to construct time-frequency spectrograms for manual feature extraction [10–12]. However, this approach imposes significant memory burdens (≈50 KB) on MCUs with limited RAM capacity (typically ≤256 KB), severely straining their already constrained memory resources. In contrast, we only perform a normalized pre-processing of the input I/Q signals, enabling the end-to-end model architecture to process input data directly into final outputs, achieving much greater computational efficiency.

3.2 CSS-informed Model Design

Necessity. Though the STFT-free processing with lightweight architectures (e.g., fully-conv, pure-GRU, and MLP) meets the efficiency requirement, we observe that their direct application to LoRa weak signal demodulation yields poor performance. This limitation stems from the model's lack of domain-specific knowledge about LoRa modulation. To address this, we need to delve into the principles of CSS modulation and integrate LoRa-specific designs into the model, enabling robust demodulation of LoRa chirp signals.

CSS-informed Convolution. The CSS modulation principle defines that each LoRa chirp has a distinct linear frequency variation pattern over time. Inspired by such fact, we propose CSS-informed convolution with a designed receptive field. Notably, an undersized convolutional receptive field would inadequately capture local frequency variations under noise, while an oversized one impairs computational efficiency. We empirically adopt a multi-scale kernel design, with sizes ranging from 1/12 of the chirp length in early layers for robust pattern detection down to 1/128 in deeper layers for fine-grained feature localization. By sliding multiple CSS-informed convolutional filters along LoRa signals, our model effectively extracts distinctive linear frequency-variation patterns that directly map to the frequency signature of the corresponding symbol. These self-learned filters develop frequency-selective capabilities: filters with high-frequency selectivity exhibit strong activation in the high-frequency regions of temporal chirp signals, while those favoring low frequencies generate weaker activations in high-frequency components. This process produces a multidimensional frequency response encoding a chirp-specific spectral pattern, demonstrating superior demodulation performance compared to traditional dechirp that relies solely on spectral energy analysis [26].

Moreover, the CSS modulation principle dictates that symbols are modulated through cyclic shifts of the base upchirp's initial frequency, making the precise detection of frequency transitions at the starting and ending points of each chirp crucial for successful demodulation. Building on this insight, we propose a boundary-aware convolutional extension strategy that deliberately shifts the convolutional window forward at the chirp's start and backward at its end, extending the receptive field beyond chirp boundaries, thereby enhancing the perception of the frequency cyclic shift.

CSSNet Structure. As illustrated in Fig. 3, CSSNet consists of an I/Q fusion block, cascaded CSS-informed feature extraction blocks and inference blocks.

I/Q Fusion Block. Such a block sequentially integrates a 1D CNN layer, a LeakyReLU activation layer, a batch normalization layer, and a dropout layer to refine I/Q signal representations. Specifically, by leveraging the inherent orthogonal I/Q signals of LoRa chirps and their shared frequency spreading patterns, this block employs multiple convolutional filters with size 1 to achieve cross-channel weighted superposition on I/Q signals through data-driven adaptive weight optimization. This process enables preliminary denoising.

CSS-informed Feature Extraction Block. Distinct from the prior block, this block further integrates a max-pooling layer and replaces the CNN with the proposed CSS-informed CNN. The enlarged receptive field enhances robustness against high noise, while max-pooling further emphasizes dominant frequency transitions and suppresses localized noise spikes. Specifically, the convolutional window for the first two blocks is set to 1/12 of the chirp length. Starting from the third module, the window size progressively decreases layer-wise down to 1/128 of the chirp length, capturing the linear frequency-variation patterns at various granularities.

Inference Block. This block serves as the final classification stage, comprising a fully connected layer, an activation layer, a dropout layer, and a softmax output. The dense layer maps chirp-specific patterns to class logits. The final softmax layer then normalizes logits into probabilistic symbol predictions.

Uplink Potential. While the computing resources of LoRa gateway chips [27] are in fact relatively limited, hindering the practicality of many PC-based academic research projects, CSSNet's lightweight design also enables deployment on COTS gateways. The structure can be strategically scaled to enhance demodulation accuracy with small memory overhead. Specifically, this enhanced variant, termed CSSNet+, features three key upgrades: (1) extending

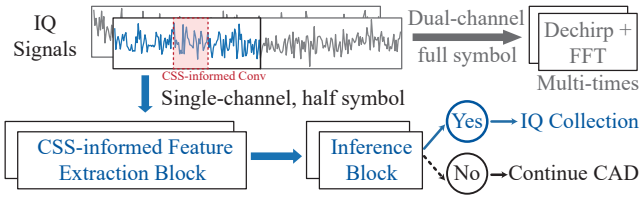


Figure 4: LoRa preamble detection process of the traditional dechirp and the proposed CADNet.

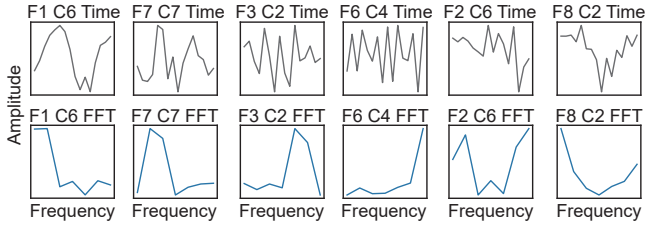


Figure 5: Illustration of convolutional filters across different channels (Conv_1d). The FFT figures demonstrate the frequency-selective behavior of the filters.

the input sampling points by 50% (via copying the first half), (2) stacking additional CSS-informed feature extraction blocks, and (3) deepening the dense layer for better discrimination.

3.3 Hardware-aligned Model Compression

Necessity. Given the memory constraints of typical MCUs in LoRa nodes, CSSNet necessitates hardware-aligned compression for efficient deployment. While post-training pruning coupled with quantization offers proven model compression benefits, the MCU's resource-limited architecture, such as lacking the support for sparse computing, imposes compatibility restrictions [23]. In this paper, we introduce MCU-aligned pruning and quantization.

Insights on MCU Architecture. Traditional unstructured pruning removes individual weights to create sparse connections [28]. However, MCU architectures lack hardware support for sparse matrix operations, forcing runtime computations to handle zero weights as active operands, which paradoxically increases clock cycles and energy consumption. Coarse-grained structured pruning represents a suitable pruning strategy for MCUs, eliminating entire filters and neurons to maintain dense computation graphs.

MCU-aligned Pruning. The TensorFlow platform possesses industry-leading support for heterogeneous MCUs, exemplified by the TensorFlow Lite framework [24, 29]. However, TensorFlow currently lacks native support for filter-level pruning algorithms. To address this, we implement such a pruning algorithm from scratch through low-level matrix operations using NumPy. Specifically, we precisely reconstruct the pruned weight matrix corresponding to each filter and neuron across all layers, enabling both fine-grained benchmarking and parameter-specific fine-tuning. By dynamically adjusting pruning ratios of each layer, CSSNet achieves granular compression for maximizing retention of meaningful weights.

MCU-aligned Quantization. We adopt post-training 8-bit quantization for maximizing the compression ratio. Specifically, we adopt static quantization to avoid runtime float-to-int conversion overheads. By pre-computing scales for both weights and activations, we convert the entire network to INT8, achieving faster inference on MCUs via pure integer arithmetic operations.

4 FAST WEAK PREAMBLE DETECTION

The objective of the fast weak preamble detection, is to detect LoRa preambles under low SNR within ultra-short preamble durations. Specifically, such technique consists of *lightweight neural LoRa detector* (§ 4.1), and *sliced superimposition* (§ 4.2) described as follows.

4.1 Lightweight Neural LoRa Detector

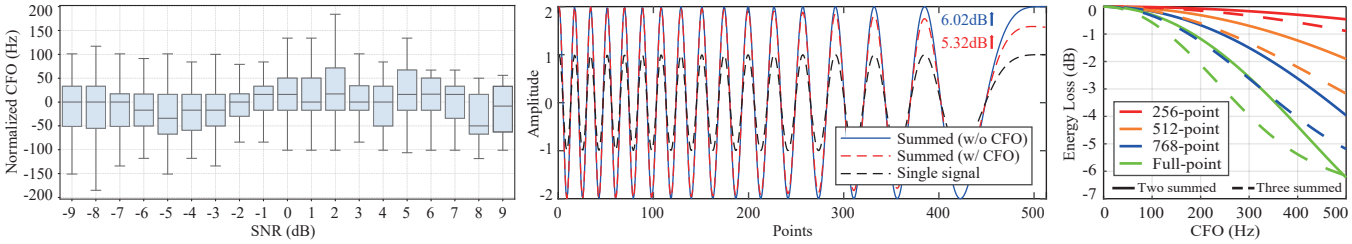
Necessity. The LoRa preamble consists of a series of base upchirps, which create a fixed pattern that the receiver can detect. On the one hand, LoRa chips detect packets typically through multiple dechirp + FFT operations, as shown in the gray part of Fig. 4, which cannot detect the signal under the standard dechirp SNR threshold. On the other hand, coherent accumulation of identical chirps can achieve noise suppression, yet requires repetitive sliding-window FFT processing [8, 9], which is computationally prohibitive for MCUs, causing critical latency. In this paper, we consider preamble detection as a simplified binary classification task rather than multi-symbol demodulation, significantly reducing the algorithmic complexity for weak preamble detection.

CADNet. Inspired by CSSNet, we design CADNet, a lightweight neural LoRa detector with a reduced number of CSS-informed blocks for basic LoRa chirp features extraction. To further minimize MCU computational overhead, we reduce the input signal dimensionality from dual-channel to single-channel by discarding the Q channel, and slice the sampling window in half. This modification slashes the input memory footprint by 75% while preserving essential temporal features. As illustrated in Fig. 4, CADNet begins with single-channel feature extraction, followed by feature mapping through the inference block. The model outputs a binary decision indicating whether a LoRa chirp is detected. If detected, the MCU activates I/Q data streaming for demodulation; otherwise, it remains in low-power preamble detection mode.

Filter Illustration. We visualized representative CSS-informed convolutional filters as shown in Fig. 5. The waveforms (blue line) depict frequency-domain values of kernels across multiple channels in the final convolutional layer, demonstrating the frequency-selective behavior of filters. Given that time-domain convolution equates to frequency-domain multiplication, these kernels effectively extract frequency variation features, which are strongly correlated with the CSS patterns inherent in LoRa chirp signals.

4.2 Sliced Superimposition

We integrate coherent superposition methods with CADNet to enhance performance. Specifically, superposition of same-sized time-domain signals is first implemented, achieving a theoretically optimal 6 dB gain if two phase-aligned signals are coherently summed ($20log_{10}(S_{\text{sum}}/S_{\text{single}})$, where S_{sum} and S_{single} denote the signal amplitudes after superposition and of a single signal). To accommodate CADNet's input configuration, single-channel half-length chirp processing, we implement a sliced superimposition mechanism, which involves temporally slicing the received single-channel chirp signals in half followed by superimposing the halves of these signals. Moreover, the summation process simultaneously functions as a spectral filtering mechanism. Only consecutively transmitted identical LoRa preambles are constructively enhanced, inherently suppressing interference signals.



(a) Normalized CFO Distribution across SNR Levels. (b) Summed Two Consecutive Single-channel Signals (CFO = 300Hz). (c) Energy Loss versus CFO.

Figure 6: Coarse CFO recovery evaluation. (a) Normalized CFO distribution across different SNR levels. It demonstrates that SNR has little influence on the CFO. (b) Two consecutive single-channel signals are summed up for energy condensing, and CFO induces energy loss. (c) Energy loss versus CFO for different signal lengths.

5 EFFICIENT OFFSET RECOVERY

The objective of the efficient offset recovery is to improve the efficiency of signal synchronization on LoRa nodes with a priori knowledge. This section consists of *efficient CFO recovery* (§ 5.1) and *other offset recovery* (§ 5.2).

5.1 Efficient CFO Recovery

Necessity. The frequency offset Δf between transmitted and received signals introduces time-accumulating phase shifts in windowed chirp signals, causing destructive interference for signal superposition and demodulation. In this paper, we propose coarse- and fine-grained CFO recovery techniques for LoRa nodes.

Coarse-grained CFO Recovery. We observe that CFO in LoRa systems primarily originates from local oscillator mismatches between transceivers of different devices, which predominantly remain stationary in practical deployments [5]. This insight reveals that historical CFO measurements provide valuable prior knowledge for current CFO estimation through averaging/median operations, enabling bounded-error coarse CFO recovery. Specifically, prior to sliced superimposition, the second signal segment is multiplied by $Re[e^{-j2\pi\Delta f \cdot T}]$, where T denotes the chirp duration.

CFO Test. We conduct an experiment where three commercial LoRa nodes with different transceiver chips are configured to send and receive signals to each other across varying distances (2s transmitting interval per node). By leveraging on-chip CFO estimation and SNR measurement capabilities, we collected over 6,000 SNR-CFO data pairs characterizing CFO variations across different LoRa nodes under varying SNR conditions. We normalized the CFO values by subtracting their median per transceiver pair, with aggregated results presented in Fig. 6(a), demonstrating the statistical stability of CFO within 200Hz across SNR variations. This evidence confirms the effectiveness of historical CFO measurements.

CFO Influence Study. Building upon this finding, we conduct MATLAB simulations to quantify CFO impacts on coherent real-valued signal superposition (Fig. 6(b)). Ideal CFO-free signal superposition achieves 6.02 dB SNR gain through constructive amplitude combining. When introducing 300 Hz CFO, a large offset value according to tests, the phase misalignment reduces the SNR gain to 5.32 dB.

Energy Loss Study. We further analyze the CFO-induced energy loss during multi-length signal superposition, with results depicted in Fig. 6(c). Colored curves correspond to varying signal lengths, while the dashed line denotes the three-windowed signal superposition effect. As CFO induces time-accumulating phase errors, larger CFO values correlate with greater energy loss post-superposition. This loss scales with signal duration, as longer data segments amplify cumulative phase misalignment. CADNet's half-length chirp input design inherently mitigates this energy degradation.

Fine-grained CFO Recovery. In the demodulation phase, we need to perform accurate CFO recovery of the LoRa I/Q signal for precise demodulation. CFO causes a linear accumulation of signal phase over time, thus the initial phase of the kth window chirp is shifted by $\phi_{(\Delta f,k)} = 2\pi \cdot \Delta f \cdot T \cdot k$, where T is the chirp duration. Based on historical CFO f_{ref} (i.e., historical CFO mean), we can construct a search space $\Omega = [f_{\text{ref}} - 2\Delta, f_{\text{ref}} + 2\Delta]$, where the boundary parameter Δ is determined by the largest deviation in historical CFO data. The fine-grained CFO recovery is achieved by solving the following phase-compensated optimization problem:

$$\Delta f = \underset{\Delta f \in [f_{\text{ref}}-2\Delta, f_{\text{ref}}+2\Delta]}{\arg\max} \left| \mathcal{F}\left\{ \sum_{k=0}^{N-1} \left(C_k[n] e^{-j\phi(\Delta f,k)} \right) \odot C^*[n] \right\} \right|$$

where N denotes the number of concatenated chirp symbols, $C_k[n]$ represents the kth time-windowed chirp segment, $C^*[n]$ represents the base downchirp, $\mathcal{F}$ represents the max value of FFT operation. Compared with the general approach [9, 10], we do not need to search in all possible Δf, optimizing the search space based on a priori knowledge, which makes efficient CFO recovery feasible. Specifically, our approach significantly reduces the computational complexity by narrowing the search space for the Δf. The general approach must search over a wide potential frequency range, Ω_{full}, with a width of W_{general} (e.g., tens of kHz). Let S be the frequency search step and M be the number of samples per chirp symbol. The complexity of a single FFT operation is $O(M \log M)$. Thus, the total computational complexity for the general approach is $C_{\text{general}} = \frac{W_{\text{general}}}{S} \cdot O(M \log M)$. In contrast, our method leverages historical data to define a much smaller search space, $\Omega = [f_{\text{ref}} - 2\Delta, f_{\text{ref}} + 2\Delta]$, with a width of $W_{\text{efficient}} = 4\Delta$. The complexity of our efficient method is therefore: $C_{\text{efficient}} = \frac{W_{\text{efficient}}}{S} \cdot O(M \log M) = \frac{4\Delta}{S} \cdot O(M \log M)$. The efficiency gain, or Complexity Reduction Factor, can be quantified as the ratio of the two complexities:

$$\text{Complexity Reduction Factor} = \frac{C_{\text{general}}}{C_{\text{efficient}}} = \frac{W_{\text{general}}}{4\Delta}$$

For a typical scenario where W_{general} might be ±10 kHz (a 20,000 Hz range) and our historical data suggests a stable CFO variation within 400 Hz (i.e., $2\Delta = 400$ Hz, making the search width $4\Delta = 800$ Hz), the computational load is reduced by a factor of 25, achieving a more efficient CFO recovery. The historical data pool can be updated

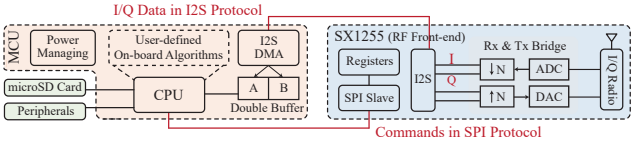


Figure 7: MCU-SDR block diagram.

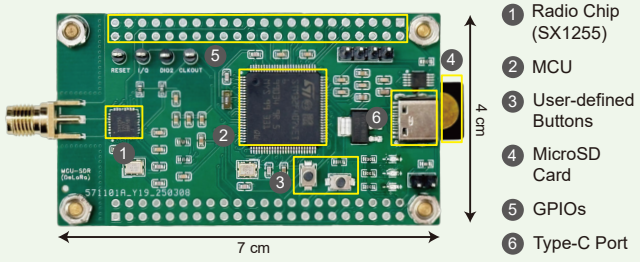


Figure 8: MCU-SDR prototype.

with the communication process, adaptively mitigating CFO drift caused by hardware aging and thermal fluctuations.

5.2 Other Offset Recovery

We need to further address the sampling time offset (STO) and time offset (TO) issues which degrade demodulation.

The time offset between signal arrival and sampling time of a receiver (i.e., STO) introduces frequency and phase distortion to received chirps. A high sampling rate helps reduce the variance of SNR gains across symbols [8, 9]. We establish through Nyquist criterion analysis that maintaining a sampling rate $F_s \geq 3BW$ is required (BW denotes LoRa bandwidth).

TO arises from sampling window misalignment. Conventional dechirp addresses TO by demodulating the LoRa preamble and aligning the FFT's energy peak to bin 0. However, dechirp fails to work under extremely low SNR. We now demodulate the preamble using CSSNet, and the result of the demodulation is written as *code*.

$$bias = \frac{F_s}{BW} \cdot (2^{SF} - code)$$

By moving the sampling window forward *bias* samples (SF is spreading factor), we can align the sampling window.

6 MCU-SDR TESTBED

Neither commercial LoRa nodes nor existing SDRs can serve as suitable testbeds for LoRa node demodulation, as the former hide baseband access behind proprietary chips and the latter rely on advanced hardware that fails to emulate computing constraints of LoRa nodes. Therefore, we developed a proper standalone SDR from the ground up, namely MCU-SDR, which comprises two core components: an MCU and a radio chip. The block diagram of MCU-SDR is shown in Fig. 7, and the prototype is shown in Fig. 8. MCU-SDR serves as a viable LoRa node demodulation testbed as the MCU has comparable computing resources to LoRa chips and meanwhile supports flexible deployment of demodulation algorithms. Specifically, I/Q signals are streamed to the MCU, which runs the demodulation software to be evaluated. The MCU and radio chip collaborate through SPI and I2S protocols to achieve highly-flexible programmability. Tunable parameters for the RF section include frequency band (400-510 MHz), sampling rate (0.2-2 MHz), LNA gain (-48 to 0 dB), and I/Q bit width (8 to 32 bits), among others.

Table 1: Comparison between different standalone SDR platforms. Data of other SDRs comes from [18, 19].

Platform	Sleep Power (mW)	Cost ($)	ADC (bits)	HW-matched	Push-the-button
USRP E310	2820	3000	12	–	–
bladeRF 2.0	717	720	12	–	–
μSDR	320	150	8	–	–
GalioT (RTL-SDR)	350	60	8	–	–
TinySDR	0.03	55	13	–	–
MCU-SDR	**0.01**	**9**	**13**	√	√

6.1 MCU-SDR Core Designs

Radio Chip Selection. Our primary requirement for the radio chip is the capability to transfer LoRa baseband I/Q signals via digital protocols natively supported by the MCU. While most existing solutions rely on LVDS protocol (e.g., AD9361) or high-speed USB 2.0 protocol (e.g., RTL2832), which lack native MCU compatibility, we surprisingly identified a subset of chips (e.g., NAU8822) transferring I/Q signals via I2S protocol, a serial bus protocol supported by most MCUs. However, these alternatives operate in lower frequency bands that are unsuitable for LoRa. After extensive investigation, we select SX1255 [20], which inherits the same RF front-end design as that of most COTS LoRa chips [16]. Unlike LoRa chips, SX1255 exposes its digital I/Q interface via I2S protocol, allowing seamless MCU interfacing. This choice enables MCU-SDR to faithfully emulate the RF behavior of LoRa nodes while retaining full programmability.

Radio Chip Configuration. Due to the absence of open-source reference code and commercial development boards for the SX1255, we designed a custom test board for firmware development and validation. Compared to the prototype shown in Fig. 8, this test board separates the radio front-end from the MCU, allowing flexible evaluation across different MCU families. We reverse-engineered the behavior of the SX1255 by decoding I2S waveforms on the oscilloscope and matching them to internal register effects. Through extensive probing, we self-developed firmware for SX1255 from scratch to configure all 19 8-bit registers. This effort lays the foundation for MCU-SDR's fully reconfigurable control of RF parameters.

CPU-less I/Q Streaming. Due to the lack of open-source implementations for MCU-based I/Q streaming, we developed an optimized MCU configuration to achieve continuous, high-throughput data transfer. The MCU operates in DMA mode, where the DMA controller adheres to the I2S protocol and utilizes a dedicated bus to directly capture I/Q signals from the SX1255 without CPU intervention. This design offloads low-level data movement from the CPU, allowing it to focus entirely on higher-level demodulation tasks. To optimize real-time processing, we employ a double-buffer (A/B) mechanism, as shown in Fig. 7. As soon as the DMA fills up one buffer, the DMA triggers an interrupt to the CPU, and then the CPU processes the I/Q signal data in that buffer, while the DMA simultaneously transfers the incoming I/Q stream into another buffer. This pipeline prevents tearing or overwrite conflicts by ensuring that the CPU only processes completed buffers, effectively decoupling data acquisition and computation for seamless I/Q streaming.

Memory Interface. During the DMA workflow, we can configure the MCU to transfer the I/Q signal from memory to the microSD card in accordance with the interrupt response sequence, which empowers us to collect enough I/Q data and analyze the spectrum.

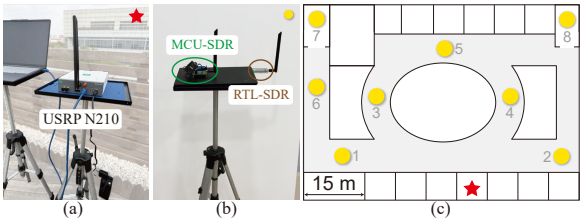


Figure 9: Indoor implementation of DeLoRa. (a) USRP N210 based gateway. (b) Rx node setup, comprising the MCU-SDR and COTS RTL-SDR (ground truth). (c) Indoor testbed.

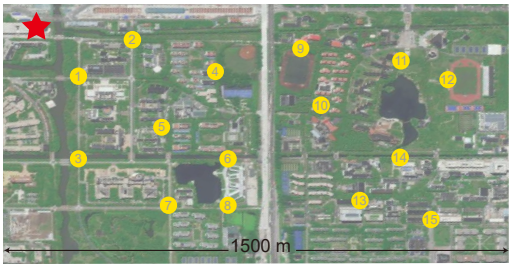


Figure 10: Campus-scale outdoor testbed.

Push-the-Button I/Q Acquisition. To facilitate easy field experiments, we integrated all the aforementioned processes into a unified firmware routine that can be triggered by an on-board button. A single press starts real-time I/Q acquisition and sequential data storage to the microSD card, while a subsequent press safely terminates the recording process.

6.2 Remarks on MCU-SDR

The comparison between MCU-SDR and other standalone SDR platforms is listed in Table 1. Benefiting from the minimal hardware combination, the sleep power and cost of MCU-SDR outperform other SDRs. Combined with a user-friendly push-the-button driver, MCU-SDR successfully democratizes LoRa downlink research. Moreover, MCU-SDR provides multiple analog and digital GPIO interfaces for peripheral integration (e.g., sensors, displays, buttons).

7 IMPLEMENTATION

7.1 LoRa Symbol Dataset

To ensure a reproducible evaluation, we utilized the well-maintained NELoRa dataset [10] for model training. This also enables a fair benchmark against the leading uplink method (NELoRa) to validate the downlink demodulation performance of CSSNet. Specifically, the dataset comprises LoRa symbols sampled in real-world environments under high SNR conditions with SF ranging from 7 to 10 and BW of 125 kHz. Each symbol is composed of dual-channel I/Q sequential signals, sampled at 1 MS/s.

7.2 Model Training and Testing

To enhance the generalization of CSSNet, we performed dataset augmentation by incorporating Additive White Gaussian Noise with varying SNR levels, a technique widely adopted in mainstream works [9–12, 30], while the synthetic dataset also facilitates fine-grained SNR control for comprehensive evaluation. For CADNet, more aggressive data augmentation is adopted to prevent overfitting. Specifically, we construct a positive-class dataset with all LoRa symbols and a negative-class dataset containing single-tone signals with different phases, BFSK signals, and pure noise signals to strengthen CADNet's perception of the CSS pattern. We then follow the same noise injection methodology as CSSNet, while maintaining a higher proportion of low-SNR data in the total dataset. For both models, we adopted an 80-20 split between training and testing datasets. We trained the models for 2000 epochs each using the AdamW optimizer with a learning rate of 3e-4.

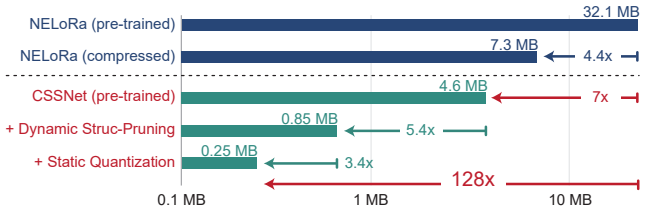


Figure 11: Model size comparison between the proposed CSSNet and baseline (NELoRa).

7.3 Methods for Comparison

We compare CSSNet with the following methods:

Highest FFT Bin (HFFT): This method is based on dechirp and demodulates symbols by identifying the position of the highest-magnitude FFT bin.

LoRaPHY [31]: This method is also based on dechirp but distinctively introduces a phase search step to coherently combine the two dispersed energy bins in the FFT spectrum.

NELoRa [10]: NELoRa is the leading neural LoRa demodulator employing dual neural networks for chirp recognition. Like other weak signal demodulation methods, NELoRa incurs substantial computing cost that hinders its deployment on LoRa nodes. We conduct a systematic comparison between CSSNet and NELoRa to demonstrate our methodological advancements.

7.4 Hardware Implementation

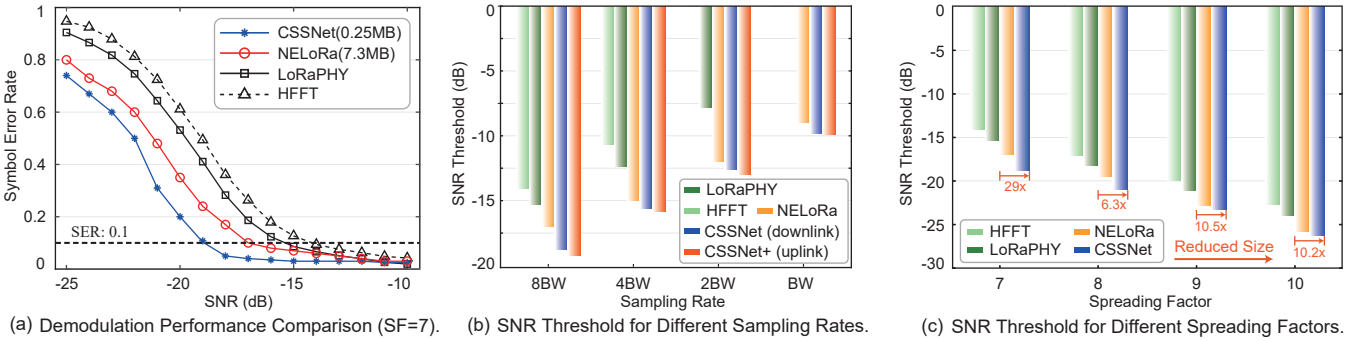
We deploy the testbed in both indoor and outdoor scenarios for validating the robustness of DeLoRa, as shown in Fig. 9 and Fig. 10. We use USRP N210 as a gateway to transmit LoRa packets based on an open-source project gr-lora [32]. Specifically, we configure LoRa packets with default parameters as: central frequency 433 MHz, spreading factor 7, bandwidth 125 kHz, coding rate 4/5 and payload size 16 chirps. Then, we use the MCU-SDR to collect and in-situ demodulate I/Q signals in different places, while using the RTL-SDR connecting a laptop for collecting I/Q signals as the ground truth.

8 EVALUATION

8.1 Model Evaluation

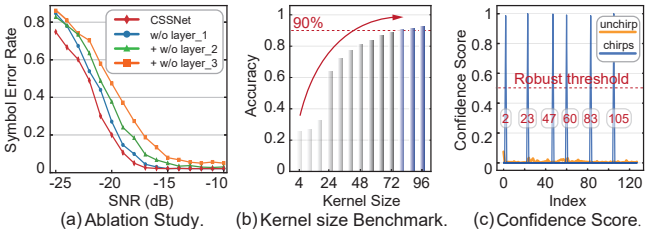
Model Compression. We compare the model size of CSSNet with the baseline in Fig. 11. Thanks to its lightweight model architecture, the pre-trained CSSNet is 7× smaller than NELoRa. After hierarchical compression including dynamic structural pruning and static quantization, CSSNet is further compressed to 0.25 MB, 128× smaller than pre-trained NELoRa, while maintaining the accuracy.

CSSNet Analysis. A detailed comparison between demodulation methods at different SNRs at SF7 is shown in Fig. 12(a), which are all based on a sampling rate of 1MS/s. We use the SNR at symbol



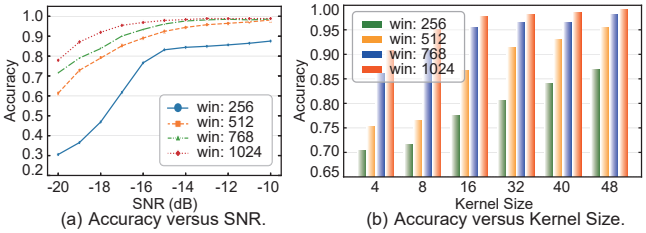
(a) Demodulation Performance Comparison (SF=7). (b) SNR Threshold for Different Sampling Rates. (c) SNR Threshold for Different Spreading Factors.

Figure 12: CSSNet performance. (a) Demodulation performance comparison between CSSNet and baselines. (b) The influence of sampling rate on the SNR threshold. (c) SNR threshold across different spreading factors.



(a) Ablation Study. (b) Kernel size Benchmark. (c) Confidence Score.

Figure 13: CSSNet benchmark. (a) Ablation study of CSSNet. (b) The influence of kernel size on SNR threshold. (c) CSSNet confidence score illustration.



(a) Accuracy versus SNR. (b) Accuracy versus Kernel Size.

Figure 14: CADNet benchmark. (a) Accuracy versus SNR. (b) Accuracy versus kernel sizes.

error rate (SER) 10% as a threshold and calculate the SNR thresholds of different methods at sampling rates from BW to 8BW (SF7) and SF from 7 to 10 (sampling rate 8BW), shown in Fig. 12(b) and (c). CSSNet achieves a 1.8 dB SNR gain over NELoRa and a 3.3 dB SNR gain over LoRaPHY at 10% SER threshold at SF7. At all sampling rates, CSSNet outperforms other methods and the extended CSSNet+ can reach an even lower SNR threshold, demonstrating the uplink potential of our method. Even under sampling rate BW (theoretically violating Nyquist constraints), CSSNet can effectively work under lower SNR, empowering resource-constrained devices to dynamically select sampling rates based on their memory conditions. While conventional demodulators fail under small sampling rates as spectral aliasing corrupts signal information, AI models succeed by learning to interpret this predictable distortion as a recognizable pattern. Across SF from 7 to 10, CSSNet reaches a lower SNR threshold than NELoRa while achieving a smaller model size, benefiting from our CSS-informed model designs.

Moreover, we conduct a comprehensive benchmark for CSSNet. Fig. 13(a) presents an ablation study where progressively disabling layers degrades demodulation accuracy, confirming the significance of the I/Q fusion layer and stacked CSS-informed feature extraction layers. Fig. 13(b) benchmarks the impact of kernel size (receptive field) in feature extraction layers, revealing that optimal kernel scaling is essential for capturing the distinct LoRa frequency variation pattern. Finally, Fig. 13(c) shows the confidence score of CSSNet while processing different LoRa chirps and noise (unchirp) under -18 dB. CSSNet maintains >90% confidence score for valid chirps versus <10% for noise, enabling a robust threshold to determine LoRa demodulation completion.

CADNet Analysis. Fig. 14(a) quantifies the accuracy of CADNet across varying input lengths and SNR conditions with a kernel size of 32, demonstrating that longer input signals enable the model to better capture LoRa chirp characteristics for preamble identification. Compared to CSSNet, CADNet achieves LoRa signal detection under lower SNR conditions while employing only sliced single-channel signals as input. Unlike traditional LoRa preamble detection methods that rely on correlation-based matching, CADNet leverages learned chirp representations, allowing reliable detection under lower SNR and even reducing latency (as will be shown in § 8.3). Fig. 14(b) illustrates the accuracy of CADNet across varying input lengths and kernel sizes at an SNR of -15 dB, revealing that larger receptive fields enhance model performance, further validating the rationality of our convolutional design. CADNet achieves robust performance with smaller receptive fields compared to CSSNet since its objective is to detect the presence of LoRa symbols rather than perform fine-grained symbol classification.

8.2 In-Situ Evaluation

MCU-SDR Test. Since the I/Q signals from the SX1255 are transmitted via the I2S protocol, we initially validate the data communication compliance with the I2S standard using an oscilloscope, confirming that the SX1255 accurately transmits I/Q signals following the I2S protocol, where the I signal is transmitted when WS is low and the Q signal is transmitted when WS is high.

Subsequently, we verify whether the MCU in the MCU-SDR correctly received I/Q signals by employing an RTL-SDR as ground truth to capture LoRa packets transmitted by a USRP. We configured the MCU-SDR to initiate I/Q collection upon LoRa packet detection and transmit the I/Q signals via the MCU's serial port to a PC. By comparing the successfully demodulated I/Q signals under high SNR conditions, we confirmed that MCU-SDR accurately aligns and streams raw I/Q signals with no sample discrepancies.

Robustness in Different Environments. We conducted experiments in both real-world indoor and outdoor environments, where the indoor setting exhibited rich multipath effects while the

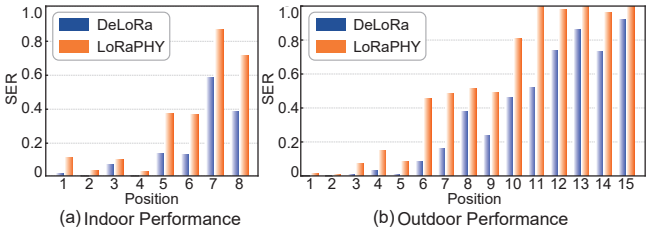


Figure 15: Cross-position performance. (a) Indoor scenario. (b) Outdoor scenario.

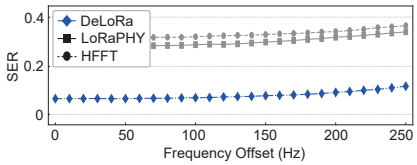


Figure 16: Performance under manually induced CFO.

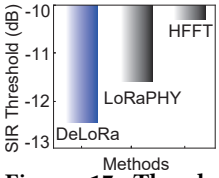


Figure 17: Threshold of SIR.

outdoor campus-scale scenario contained abundant trees, buildings, moving pedestrians, and vehicles. Specifically, we configure the indoor gateway at 0 dB transmit power to simulate LoRa signals severely attenuated by walls. We utilized the CADNet to detect LoRa packets, while taking the post-measured CFO as reference for coarse- and fine-grained CFO recovery. We used LoRaPHY as the baseline as it outperforms the HFFT method and adapts to the limited computing resources of LoRa nodes. Other advanced methods were not selected for comparison, as their relatively high computational and memory requirements make them unsuitable for deployment on LoRa nodes. MCU-SDR computed the demodulation SER for the detected symbols at all 23 test positions for in-situ evaluating DeLoRa and LoRaPHY. We also recorded I/Q samples onto the SD card for offline analysis, which is consistent with the results of the in-situ test, as shown in Fig. 15. Our approach achieved lower SER than LoRaPHY at all positions, indicating significant potential for extending the downlink communication range.

Resistance against CFO and Interference. We manually configured a USRP to transmit LoRa signals with varying CFOs, using the MCU-SDR to receive and demodulate packets in real-time to evaluate the corresponding demodulation performance of different methods without applying CFO recovery for all methods. The results are shown in Fig. 16, which indicates DeLoRa outperforms other methods. In an indoor environment, we set up a USRP to transmit LoRa packets while configuring another USRP to generate continuous single-tone interference in the same frequency band. We then leveraged the MCU-SDR to test the performance of different methods under varying signal-to-interference ratio (SIR) conditions. The SIR thresholds are illustrated in Fig. 17, demonstrating the anti-interference ability of DeLoRa.

8.3 Computational and Memory Efficiency

The memory footprint of algorithms during actual operation is critical to their practicality. Table 2 presents the storage overhead of different methods for processing a SF7 symbol. Compared to leading approaches, CSSNet and CADNet require minimal RAM and FLASH space, making them perfectly compatible with the constrained memory resources of MCUs (RAM: 256 KB, Flash: 1 MB). Table 3 details the runtime latency of LoRaPHY and two

Table 2: Storage overhead for different methods.

Method	Peak RAM Size (KB)	Used Flash Size (KB)
NELoRa	N/A	7300
LoRaTrimmer	>1024	N/A
CSSNet	**28.93**	**219.17**
CADNet	**3.97**	**2.57**

Table 3: Computing time for processing a chirp.

Computing Unit	LoRaPHY (ms)	CADNet (ms)	CSSNet (s)
STM32F7 (<$3)	2.53	0.72	0.14
STM32F4 (<$1)	3.85	1.27	0.31
STM32L4 (<$1)	8.30	1.96	0.58
PC (Thinkbook)	0.053	<0.001	0.00039

proposed models for processing a SF7 symbol across various MCU platforms and a PC (CPU: Ultra 9-185H, GPU: RTX4060). To ensure performance, we selected an upsampling factor of 10 for LoRaPHY and leveraged the MCU's Floating Point Unit to accelerate the FFT computations. CADNet exhibits significantly lower latency than LoRaPHY. For the PC, CSSNet achieves a runtime latency of even less than 1 ms. While CSSNet's latency on MCUs appears high, the energy cost remains acceptable for typical low-duty-cycle IoT scenarios. For instance, most smart meters communicate with gateways only a few times per day to extend battery life [5, 33, 34], which means DeLoRa incurs negligible long-term energy overhead, as the device enters sleep mode most of the time. The total energy consumption E over a period T can be expressed as:

$$E = V \times (I_{work} \times T_{process} + I_{sleep} \times (T - T_{process}))$$

where V denotes the supply voltage, while I_{work} and I_{sleep} represent the working and sleeping current, respectively. The total processing time is $T_{process} = N_{chirp} \times t_{process}$, where N_{chirp} is the number of chirps in the payload and $t_{process}$ is the processing time per chirp. We consider a typical scenario with $N_{chirp} = 16$ (typical for ACK packets with minimal payload) and a communication interval of $T =$ 8 hours (3 times per day). By measuring the I_{sleep}^{MCU} ($\approx$35 μA), $t_{process}^{Dechirp}$ ($\approx$1 ms for SF7) and I_{work}^{MCU} ($\approx$12 mA) of the LoRa chip and MCU using a current analyzer, calculations show that DeLoRa maintains mW-level power consumption during working and overall introduces less than 6% additional energy cost to LoRa nodes, while potentially offsetting this cost through reduced retransmission energy due to enhanced downlink reliability. This energy overhead is negligible compared to the 3.3 dB enhancement, which corresponds to a 1.58× increase in communication range as validated in [35].

9 RELATED WORK

LoRa Weak Signal Demodulation. There is an increasing focus on demodulating LoRa weak signals. MALoRa [7] enhances reception by coherently combining signals from multiple antennas, while the recent MoLoRa [13] leverages intelligent antenna mobility. Instead of heavy hardware adjustments, XCopy [8] improves signal energy through coherent combination of retransmissions on the algorithmic side. LoRaTrimmer [9] optimizes energy condensation via dynamically trimming FFT windows, whereas Canas [14] recovers weak signals by canceling strong interference. With the development of AI, more and more neural-enhanced studies have emerged to improve LoRa demodulation. NELoRa [10] proposes a dual-DNN pipeline combining signal denoising and classification, while ChirpTransformer [11] introduces novel LoRa modulation

schemes demodulated by a neural network to expand spectral efficiency. Furthermore, GLoRiPHY [12] employs a transformer-based architecture, which is further advanced by LoRaSeek [15] through hierarchical feature extraction. However, the high computing and memory requirements of these methods hinder deployment on resource-constrained LoRa nodes. DeLoRa is the first method for demodulating weak signal on ubiquitous LoRa nodes.

SDR Platform. Communication research critically relies on software-defined testbeds to deploy and validate advanced methodologies. Current experimental platforms are predominantly based on high-end SDRs such as USRP N210, aligning with practical uplink scenarios due to their computing capacity mirroring resource-rich gateways, rendering them incompatible with LoRa nodes that are resource-constrained. While low-cost commercial SDRs (e.g., RTL-SDR, SDRPlay [36], LimeSDR [37]) reduce expenses, they require external PCs for computing. Similarly, standalone SDRs (e.g., bladeRF [38], μSDR [18], TinySDR [19]) use on-board FPGA as the computing hardware, mismatching with resource-limited LoRa nodes. To the best of our knowledge, MCU-SDR is the first LoRa node testbed that enables in-situ validation of demodulation algorithms.

10 CONCLUSION AND DISCUSSION

This paper proposes DeLoRa, the first weak signal demodulation scheme for LoRa nodes, breaking the SNR threshold of traditional demodulation methods. In addition, we present MCU-SDR, the first low-cost and standalone SDR without FPGA for in-situ evaluating LoRa node demodulation algorithms. DeLoRa integrates three key components: a CSS-informed neural demodulator, a weak preamble detector, and an efficient CFO recovery technique. Experimental results demonstrate that DeLoRa delivers a 1.8 dB SNR gain over the leading neural-enhanced baseline runnable only on gateways and a 3.3 dB SNR improvement over standard LoRa.

While DeLoRa achieves real-time packet detection, its demodulation of low-SNR LoRa packets, like most academic solutions [10–12], operates in a store-then-process manner rather than in strict real time. Given LoRa's inherently low duty cycle in many applications, this demodulation approach remains acceptable [34].

LoRaWAN's network capacity notably decreases with increasing SF under fixed BW due to the stringent duty cycle constraint imposed on gateways [39, 40]. A downlink demodulation enhancement technique for the small SF like DeLoRa can improve receiver sensitivity while largely maintaining the large network capacity.

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